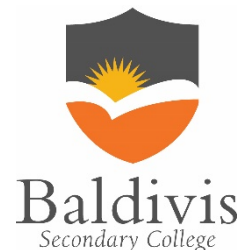


Year 9 Mini-Test 3

Probability and Linear Relationships

Resources allowed
Calculator and a page of Notes



Name: _____ Date: _____ Score: _____/40

1). If a 6 sided dice is rolled, the chance of getting a 5 is? [1 Mark]

- a) $\frac{1}{6}$ b) $\frac{5}{6}$ c) $\frac{6}{5}$ d) 1:6

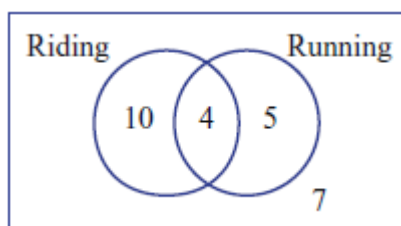
2). If a 6 sided dice is rolled, what is the chance of getting less than 4? [1 Mark]

- a) $\frac{1}{6}$ b) $\frac{1}{2}$ c) $\frac{2}{3}$ d) 4:6

3). If a 6 sided dice is rolled, what is the chance of getting the complement of rolling a 4? [1 Mark]

- b) $\frac{1}{6}$ b) $\frac{5}{6}$ c) $\frac{2}{3}$ d) 4:6

Use the following Venn diagram to complete the next 4 questions



4). How many people like running? [1 Mark]

- a) 7 b) 9 c) 10 d) 14 e) 4 f) 26

5). How many people like riding but don't like running? [1 Mark]

- a) 7 b) 9 c) 10 d) 14 e) 4 f) 26

6). How many like both running and riding? [1 Mark]

- a) 7 b) 9 c) 10 d) 14 e) 4 f) 26

7). How many people in total were asked if they like running or riding? [1 Mark]

- a) 7 b) 9 c) 10 d) 14 e) 4 f) 26

8). A family has an even chance of owning a red car, a white car or a black car. At the same time each family has an even chance of owning a dog, a cat or no pets.

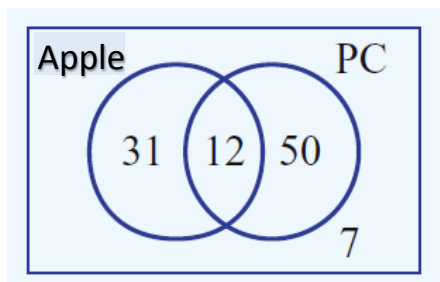
a. Draw a tree diagram. [4 marks]

b. What is the sample space of possible outcomes? [1 Mark]

c. What is the probability the family owns a white car and a dog? [1 Mark]

d. What is the probability the family has a pet and a black car? [1 Mark]

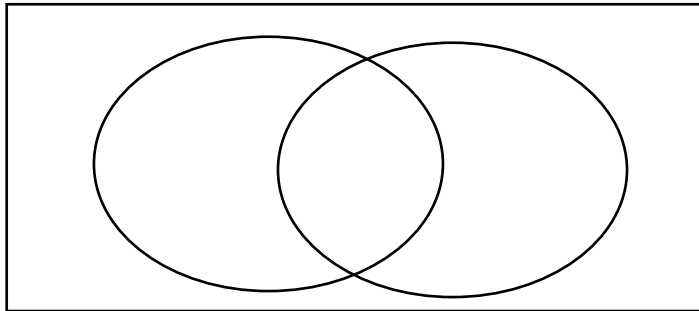
9. The Venn diagram below shows how many people own a PC computer, an Apple or both. Use this to complete the 2 way table. [3 Marks]



	Apple	No Apple	Total
PC			
No PC			
Total			

10. In a class of 30 students, 22 have an Xbox and 9 have a PS4 console. Three had both an Xbox and a PS4 console and 2 students carried neither.

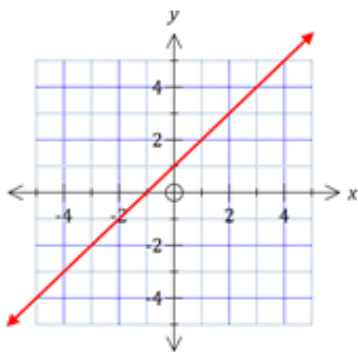
a) Represent this using a Venn diagram. [3 marks]



b) What is the probability that a student has both an Xbox and a PS4? [2 Mark]

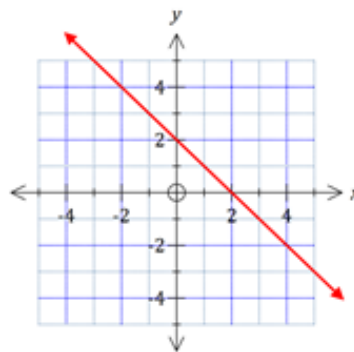
11. Find the gradient for each of the following. [1 Mark each = 3 marks]

A)

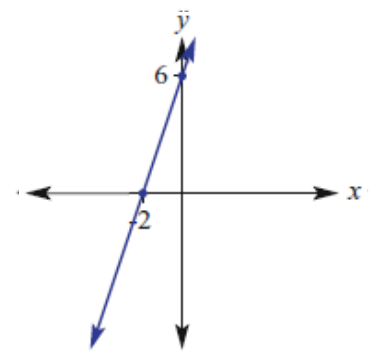


Gradient = _____

B)



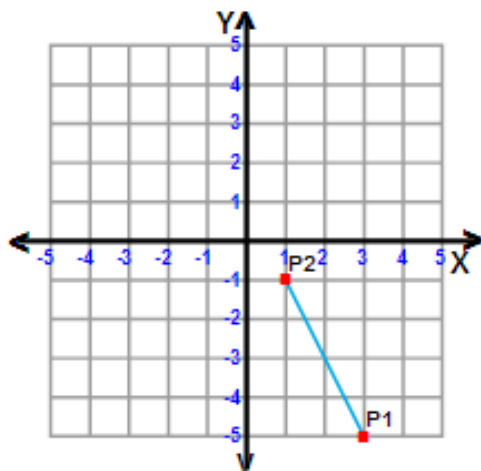
Gradient = _____



Gradient = _____

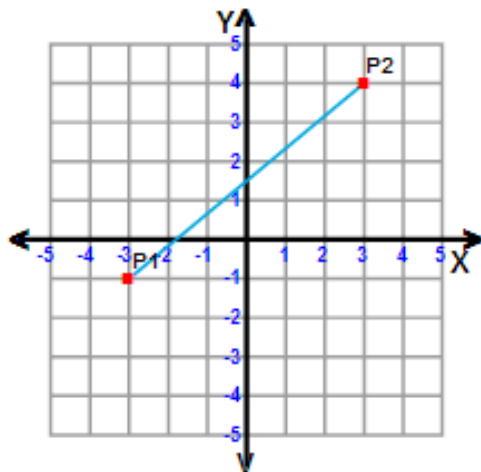
12. Find the length of the line and the coordinates for the mid point of the line for both of the following graphs [6 marks]

a)



Length of the line =

Midpoint =



Length of the line =

Midpoint =

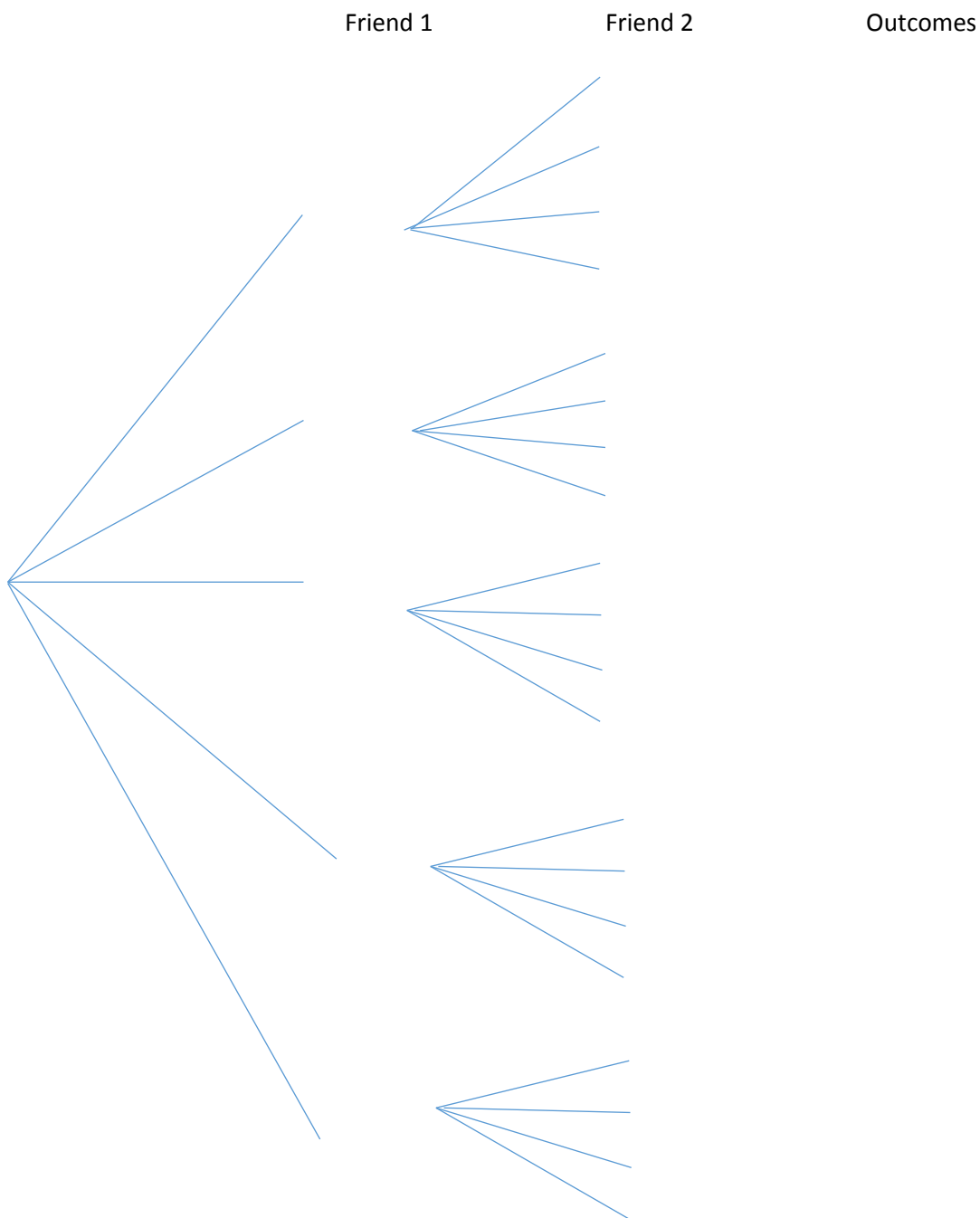
13. A survey of 25 000 new car buyers found that 750 had a major mechanical problem in the first year of operation. Calculate the relative frequency of:

a) having mechanical problems in the first year [1 mark]

b) not having mechanical problems in the first year [1 mark]

14. A friend has 5 snake lollies left in a bag, 2 green, 1 red and 2 yellow ones. He offers one snake each to his 2 friends.

a) Draw a tree diagram and list the possible outcomes of the possible choices for the 2 friends. [5 marks]



If the first friend actually takes and eats 2 snakes, 1 green and 1 yellow, what are the chances of the second friend getting the red snake? [2 mark]

End of Test